

Code Quality, Readability and Reusability Report

Smart-Lender Project Documentation Suite

Project Name	Smart Lender - AI-Powered Loan Approval Prediction System
Document Type	Code Quality, Readability and Reusability Report
Project Phase	Phase 5: Project Development Phase
Submitted By	Team Leader: Palli Dharani Team Members: Yarabarla Manidweep Saikiran Kudipudi Sita Rama Raju Datla Thota Leela Sai Krishna
Organization / College	Vishnu Institute of Technology
Status / Version	Approved / v1.0 (Exact Replica Format)

1. Production Structure Standards

Parameters Checked	Evaluation findings in Smart-Lender	Compliance Status
PEP-8 Formatting	Indentation is uniform (4 spaces). Clear spacing between classes and methods	Compliant
Naming Standards	snake_case used for variables (e.g. coapplicant_income) and camelCase for UI elements	Compliant
Comments	Detailed inline comments detailing data flow conversions and arrays shape mapping	Compliant
Error Controls	Model loading is wrapped in exception logging blocks to handle missing.pkl files	Compliant

2. Reusable Modules and Assets

Component Filename	Function Role	Reusability Scope
label_encoders.pkl	Standardized dictionary maps converting categorical user choices to integers	Used by training and application modules
scaler.pkl	Pre-calculated mean/scale values for normalization	Applies to any model deployment pipeline